Do Ragged Volume-Footprint Transformers Beat Aggregate Linear Models? A Negative BTCUSDT Development Study

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Abstract

Executed-volume footprints preserve buy and sell volume at every traded price within a bar, creating ragged sets that fixed aggregate features do not represent. We ask whether a compact hierarchical transformer that consumes all accepted rows improves direct multi-horizon forecasting over aggregate controls. FDT-N uses four learned row queries, gated fusion with 23 causal bar features, three masked temporal blocks, and eight horizon queries to predict q10, q50, and q90 marginals for seven future footprint variables. Its output parameterization prevents quantile crossing and orders the low, point-of-control, and high marginals.

We evaluate three declared seeds on 14 contiguous days of public Binance BTCUSDT individual trades: 20,160 one-minute bars, 357,676 encoded rows, and 20,089 valid 64+8 windows. A purged chronological split leaves 2,942 test windows. Across seeds, pooled normalized q50-forecast mean absolute error is 0.23643 +/- 0.00072. The validation-selected run records 0.23587, 7.25% below zero-change persistence but 2.68% above a train-only direct ridge model. Ridge has lower error in 55 of 56 horizon-target cells, and no paired four-hour block-resampling interval favors FDT-N over ridge. Pooled central-80% marginal coverage is 78.98%, but cell values range from 74.75% to 85.66% and no separate calibration set exists.

These findings do not identify a benefit from full footprint rows and do not support deployment or trading value. The defensible contribution is a causal ragged-set formulation, a physically constrained marginal-output contract, a hash-bound artifact trail, and an openly reported negative development result. Independent periods, task-matched probabilistic baselines, and controlled ablations are required.

1. Introduction

A conventional open-high-low-close-volume bar records the boundary and total activity of a trading interval. A volume-footprint bar retains a more detailed object: executed buy-initiated and sell-initiated volume at each traded price row, together with row delta, imbalance flags, point of control (POC), value area, side-specific POCs, centroids, total volume, cumulative delta, feed quality, and time. The object is attractive for market-microstructure research because it represents the realized distribution of trading activity within a bar. It is also awkward for standard forecasting: the number of price rows changes from bar to bar, the rows must remain causally sealed, and the forecast should respect both probabilistic ordering and elementary candle geometry.

Limit-order-book studies have shown that local price-volume structure can be useful input to neural models [11-14]. Their usual object, however, is displayed liquidity at fixed book levels, often paired with a future mid-price-direction label. A footprint is executed flow on a variable price support. Treating those modalities as interchangeable would obscure both the data-generating process and the meaning of a forecast.

This paper studies the following narrower question: can a compact transformer consume the complete accepted row set of sealed footprint bars and issue direct multi-horizon marginal quantiles for future footprint-derived variables? We introduce FDT-N (Footprint Distribution Transformer for next-N bars) and make four contributions.

1. We define an exact-row, causal forecast task with explicit coordinate, continuity, clipping, label, and rejection rules.
2. We combine set-style learned-query pooling [2,3] with a short causal temporal transformer [1], avoiding quadratic row-to-row attention and autoregressive forecast feedback.
3. We constrain the q10/q50/q90 outputs so quantiles do not cross and predicted price marginals obey low <= POCs <= high at each quantile.
4. We bind data, splits, runs, checkpoint, ONNX graph, runtime parity, and the manuscript to hashes, while reporting a negative comparison with a stronger linear control.

The fourth contribution is central to interpretation. FDT-N reduces pooled q50-forecast MAE relative to persistence, principally through delta and relative-volume targets, but it does not outperform the train-only direct ridge model. No experiment isolates the value of full rows, learned queries, gated fusion, or the temporal transformer. The results therefore support implementation feasibility and a reproducible experimental object, not a claim that architectural complexity improved forecasting.

2. Related work

2.1 Set-valued and ragged inputs

Deep Sets characterizes permutation-invariant functions through elementwise encoding and symmetric aggregation [2]. Set Transformer replaces fixed pooling with learned attention blocks and inducing or seed vectors [3], while Perceiver uses cross-attention into a learned latent bottleneck [23]. FDT-N adopts the same broad latent-query principle for a different statistical object: each bar is a set of executed price rows. Four learned queries attend once to the row embeddings, so attention-score count grows linearly with the number of rows. The model does not assert that the learned queries have human-interpretable footprint roles; attention weights alone are insufficient explanation [22].

2.2 Multi-horizon time-series forecasting

Temporal Fusion Transformer combines recurrent processing, attention, gating, and interpretable multi-horizon outputs [6]. DLinear demonstrates that simple linear models can be difficult to beat in long-horizon benchmarks [7], a warning directly reflected in our ridge result. PatchTST tokenizes univariate patches [8], iTransformer attends across variables [9], and Moirai trains a universal probabilistic forecaster across frequencies and variate counts [10]. These models motivate strong comparators, but none natively defines a bar as a variable-size set of executed price rows. We do not claim superiority over them; no matched adaptation was trained in this study.

2.3 Market-microstructure models

DeepLOB uses convolutional structure followed by temporal modeling on fixed-depth limit-order-book states [11]. Transformer-based LOB models likewise operate on displayed levels or engineered book features [12]. The FI-2010 benchmark formalized fixed-level mid-price prediction [13], while deep order-flow-imbalance research has emphasized the strength of comparatively simple structural signals at multiple horizons [14]. These results motivate spatial-before-temporal processing and strong linear controls. They do not validate forecasts of executed-volume footprints, whose support, targets, and measurement process differ.

2.4 Probabilistic evaluation under dependence

Quantile regression estimates conditional quantiles using the asymmetric pinball loss [15]. Proper scoring rules reward honest probabilistic forecasts in expectation [16], while conformalized quantile regression can adjust interval coverage when a suitable calibration design and assumptions are available [17]. Financial forecast errors are serially dependent and overlapping horizons share realizations. We therefore report paired block-resampling sensitivity summaries inspired by dependent bootstrap methods [18], but we do not label them formal confidence intervals: only 13 blocks are available, adjacent boundary windows overlap, and no multiplicity correction was applied. A later confirmatory study should predeclare a dependence model and may use loss-differential procedures such as Diebold-Mariano with explicit horizon and multiplicity handling [19].

3. Forecasting task and data

3.1 Sealed footprint bars

For bar t and price row j, let p[t,j] be the snapped price, v+[t,j] buy-initiated volume, and v-[t,j] sell-initiated volume. Row volume and delta are

$$v_{t,j} = v^{+}_{t,j} + v^{-}_{t,j}, \qquad \Delta_{t,j} = v^{+}_{t,j} - v^{-}_{t,j}.$$

A bar is sealed only after its interval ends. All row collections, aggregate features, references, and quality fields used at cutoff t are functions of bars at or before t. A training window is valid only when every timestamp difference across the 64 context bars and eight label bars equals exactly one minute. Missing or irregular intervals are rejected rather than treated as adjacent market time.

Given lookback L=64 and horizon H=8, the model estimates

$$\widehat{\mathbf{Y}}_{t+1:t+H} = F_\theta(\mathcal{B}_{t-L+1:t}),$$

where each bar B contains its ragged row set and the causal bar vector. The output tensor has shape [batch, 8, 7, 3]: eight horizons, seven target families, and three marginal quantiles.

3.2 Inputs

Each accepted row contributes 11 features: price offset from the bar POC; buy, sell, and total shares; delta share; log buy and sell ratios; bid and ask imbalance flags; and zero-bid and zero-ask flags. Price offsets are expressed in configured footprint-row increments. The encoded series stores all rows in one flat array plus bar offsets. Windows and batches pad only to their widest member, and masks exclude padding from attention.

Each bar contributes 23 causal features: total POC relative to the cutoff anchor; buy and sell POC offsets; total, buy, and sell centroids; low, high, and value-area gaps; relative log volume; buy and sell fractions; delta fraction; context-relative cumulative-delta change; stacked-buy and stacked-sell fractions; row-count fraction; feed quality; and sine/cosine encodings of minute-of-day and weekday. The price anchor is the last context POC. The volume reference is the median of log(1+volume) over at most the final 16 context bars.

Rows are neither truncated nor compacted for accepted bars. A bar with more than 1,024 rows is rejected. The observed maximum is 605, so this corpus has no row-bound rejection. This is an exact-row-under-a-bound policy, not an unbounded-input claim.

3.3 Targets

Five price targets - total POC, low, high, buy POC, and sell POC - are direct offsets from the last context POC, normalized by 64 footprint-row increments and clipped to [-2,2]. Volume is the future log-volume difference from the context-only reference, scaled by 4 and clipped to [-2,2]. Delta is the future buy-minus-sell fraction in [-1,1]. Side-POC cells are masked when the corresponding side has no volume. Direct anchor-relative targets avoid reconstructing an absolute forecast by adding two unrelated predicted marginals.

3.4 Frozen development corpus

The data source is the public Binance Data Vision BTCUSDT spot individual-trades archive [20]. Fourteen daily archives from 1-14 January 2024 were aggregated into one-minute bars using a 2.5 USDT footprint row. Quantities follow the terminal ingestion coordinate: quantity times 1,000, nearest-integer ties-to-even rounding, and zero-size removal. The manifest records each archive name, original byte count, and SHA-256. Raw archives are publicly reacquirable; local copies were removed after encoding.

Corpus property	Value
Instrument and venue	BTCUSDT spot, Binance
Period	2024-01-01 00:00 to 2024-01-14 23:59 UTC
Bar interval / footprint row	60 seconds / 2.5 USDT
Bars / encoded price rows	20,160 / 357,676
Rows per bar, p50 / p95 / p99 / maximum	13 / 47 / 94.41 / 605
Valid 64+8 windows	20,089
Train / validation / test windows	13,990 / 2,869 / 2,942
Purge at each split boundary	72 bars

The split is chronological: 70% train, 15% validation, and 15% test before two-sided purges equal to lookback plus horizon. The test contexts run from 12 January 2024 22:50 UTC through 14 January 2024 23:51 UTC. There is no separate calibration partition or untouched outer period.

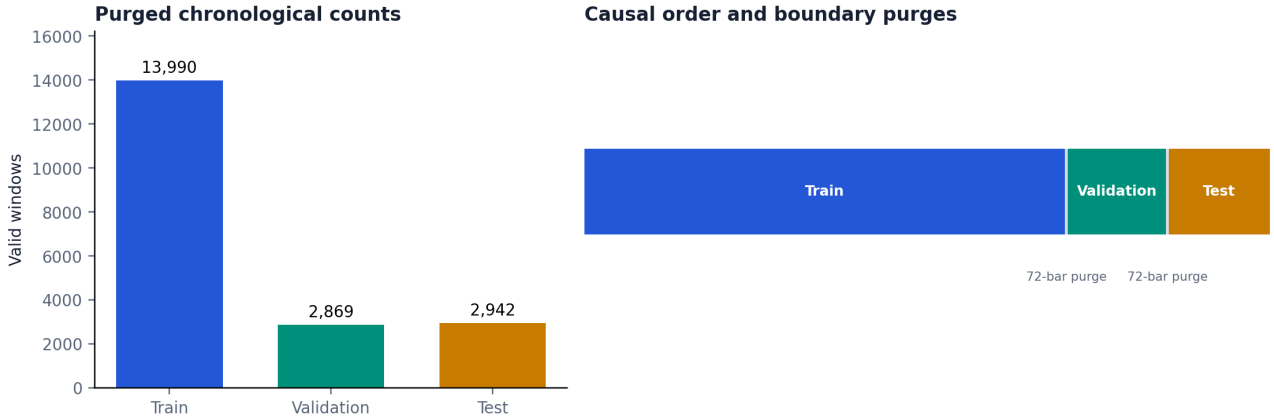


Figure 1. Frozen corpus and purged chronological split. Counts refer to valid 64-bar-context plus 8-bar-label windows after continuity screening.

4. FDT-N architecture

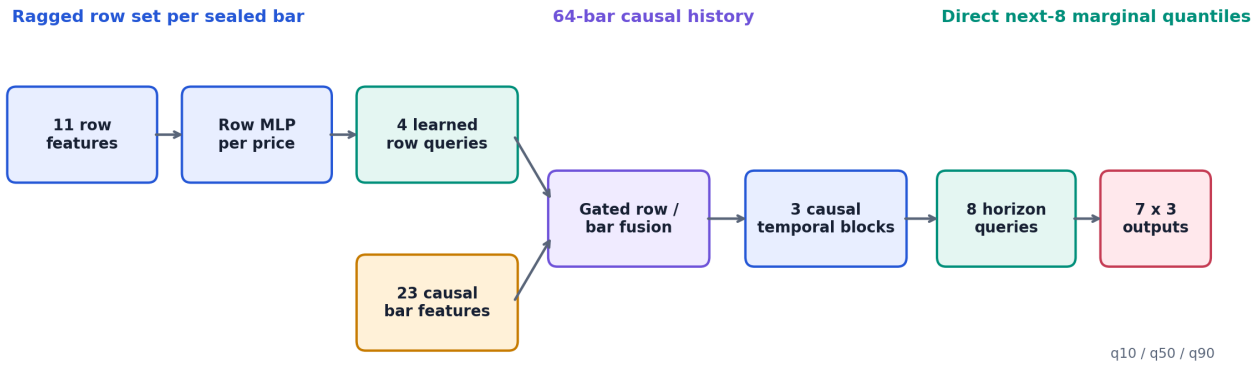


Figure 2. FDT-N architecture. Learned row queries summarize each variable-size executed-volume row set; gated bar tokens pass through causal temporal blocks, and parallel horizon queries decode all eight futures without feeding predictions back into the model.

4.1 Row encoder and learned-query pooling

Each 11-dimensional row vector x[t,j] is independently projected to model width d=64:

$$e_{t,j} = \mathrm{RMSNorm}\big(W_2\,\mathrm{GELU}(W_1 x_{t,j} + b_1) + b_2\big).$$

RMS normalization follows Zhang and Sennrich [4]. Four learned query vectors attend to the row embeddings. For query k and one attention head,

$$\alpha_{t,k,j} = \mathrm{softmax}_j\left(\frac{(W_Q q_k)^\top (W_K e_{t,j})}{\sqrt{d_h}} + m_{t,j}\right), \qquad a_{t,k} = \sum_j \alpha_{t,k,j} W_V e_{t,j},$$

where m masks padded rows. With K=4 queries and n[t] rows, the attention-score term is O(K n[t]) rather than O(n[t]^2). The four outputs are concatenated and projected to one row-set vector. The queries are learned latent slots; intended labels such as total, buy, sell, and imbalance are not treated as empirical interpretations.

4.2 Gated row/aggregate fusion

Let a[t] be the projected row-set summary and b[t] the projected 23-feature bar vector. Fusion is

$$g_t = \sigma(W_g[a_t; b_t] + b_g), \qquad u_t = g_t \odot a_t + (1 - g_t) \odot b_t + W_r[a_t; b_t].$$

The gate allows coordinate-wise weighting while the residual projection avoids forcing a convex mixture. Whether this is better than concatenation, addition, or either branch alone is untested.

4.3 Causal temporal encoder and parallel horizon decoder

A learned position embedding is added to 64 fused bar tokens. Three pre-normalized blocks apply four-head masked self-attention, SwiGLU feed-forward layers [5], dropout 0.10, and LayerScale initialized at 0.01 [21]. A triangular mask prevents position i from attending to later positions even though all 64 tokens are historical at final inference; this preserves an online interpretation for intermediate states.

Eight learned horizon queries then pass through two decoder blocks. The queries may attend to one another because none contains a future realization or previous prediction. Each query cross-attends to all encoded history states. The decoder therefore estimates all horizons directly:

$$(\hat{Y}_1, \ldots, \hat{Y}_8) = F_\theta(X_{t-63:t}), \qquad \hat{Y}_h \leftarrow \hat{Y}_{1:h-1}.$$

This removes forecast-as-input error propagation; it does not make horizon errors independent.

Architecture item	Value
Model width / attention heads	64 / 4
Learned row queries	4
Temporal blocks / decoder blocks	3 / 2
Feed-forward multiplier / dropout	3 / 0.10
Lookback / forecast horizons	64 / 8 bars
Trainable parameters	371,861

4.4 Ordered marginal quantiles and physical constraints

For target k and horizon h, a head emits a center c and two positive increments produced by softplus. The ordered quantiles are

$$\hat{Q}_{0.1} = c - \mathrm{softplus}(s^-), \quad \hat{Q}_{0.5} = c, \quad \hat{Q}_{0.9} = c + \mathrm{softplus}(s^+).$$

For every quantile separately, low is the lower anchor, high is low plus a positive range, and each POC is mapped through a sigmoid into that interval. This guarantees low <= total/buy/sell POC <= high in normalized target space, a hard-parameterization alternative to post-hoc noncrossing adjustments [24]. It does not define a coherent joint candle distribution: the outputs are seven constrained marginal distributions, not sampled paths with learned dependence.

5. Training and evaluation

5.1 Objective and optimization

For quantile tau, prediction q, and target y, pinball loss is

$$\rho_\tau(y-q) = \max\{\tau(y-q), (\tau-1)(y-q)\}.$$

Loss is averaged over valid target cells and q10/q50/q90, with horizon weight proportional to 1/sqrt(h). Training uses AdamW, peak learning rate 3e-4, weight decay 1e-3, gradient clipping, warm-up followed by cosine decay, deterministic algorithms, and early stopping with patience eight. The batch item cap is 128, but row-token budgeting reduces item count around unusually wide windows without deleting rows. Three declared seeds - 17, 29, and 43 - use identical data, configuration, and parameter count.

The checkpoint with minimum validation weighted pinball loss is saved within each seed. The primary run is the seed with minimum validation loss, tie-broken by seed number. No test score, plot, baseline comparison, or runtime measure enters selection.

5.2 Baselines

Persistence predicts zero change from the context anchor for the five price offsets and uses causal context references for the two flow targets. It has no fitted parameters. Direct ridge uses the latest, context-mean, and context-trend bar aggregates to fit an independent point forecast for every horizon-target cell. Ridge penalty alpha=1 is fixed and fitting uses training windows only. Both baselines are evaluated on byte-identical test indices.

The baselines answer a limited question: whether FDT-N's q50 is better than simple causal point predictions. They do not evaluate q10/q90 or proper distributional skill. No task-matched quantile tree, aggregate-only neural model, sequence model, or adapted time-series foundation model was trained. Consequently the paper does not claim probabilistic superiority.

5.3 Metrics and dependence treatment

We report normalized pinball loss at q10/q50/q90, MAE of the q50 forecast, signed q50 bias, central-80% marginal coverage, mean q90-q10 width, and POC-offset sign agreement. The historical JSON field `median_mae_normalized` means mean absolute error of the median (q50) forecast; it is not the median of absolute errors.

For each seed and each of 56 horizon-target cells, we resample paired model-minus-baseline absolute errors using 2,000 draws over 13 contiguous blocks of 240 test examples. A cell is described as favoring FDT-N only when the complete 95% resampling interval is below zero. These are sensitivity intervals, not independent-sample confidence intervals or significance tests. Adjacent windows at block boundaries overlap; the block count is small; and no family-wise multiplicity correction is applied. No pooled effect interval is computed.

6. Results

6.1 Seed selection and stability

Seed 17 is selected solely by validation loss. Pooled q50-forecast MAE varies by 0.00137 across seeds, whereas POC-offset sign agreement ranges from 49.44% to 54.90%. The latter is a sign diagnostic for normalized POC displacement, not candle direction, return direction, or trading performance.

Seed	Best epoch	Validation loss	Test loss	Pooled q50 MAE	Central-80% coverage	POC sign agreement
17	21	0.112634	0.070392	0.235870	78.98%	54.90%
29	13	0.114391	0.071542	0.237238	78.96%	49.44%
43	20	0.113713	0.070793	0.236171	80.23%	50.71%
Mean +/- sample SD	-	0.113579 +/- 0.000886	0.070909 +/- 0.000584	0.236426 +/- 0.000719	79.39% +/- 0.73 pp	51.69% +/- 2.86 pp

The selected run's pooled q10/q50/q90 pinball losses are 0.055493, 0.117935, and 0.051387. Mean normalized interval width is 0.720113 and pooled q50 bias is +0.034807.

6.2 Point-baseline comparison by target

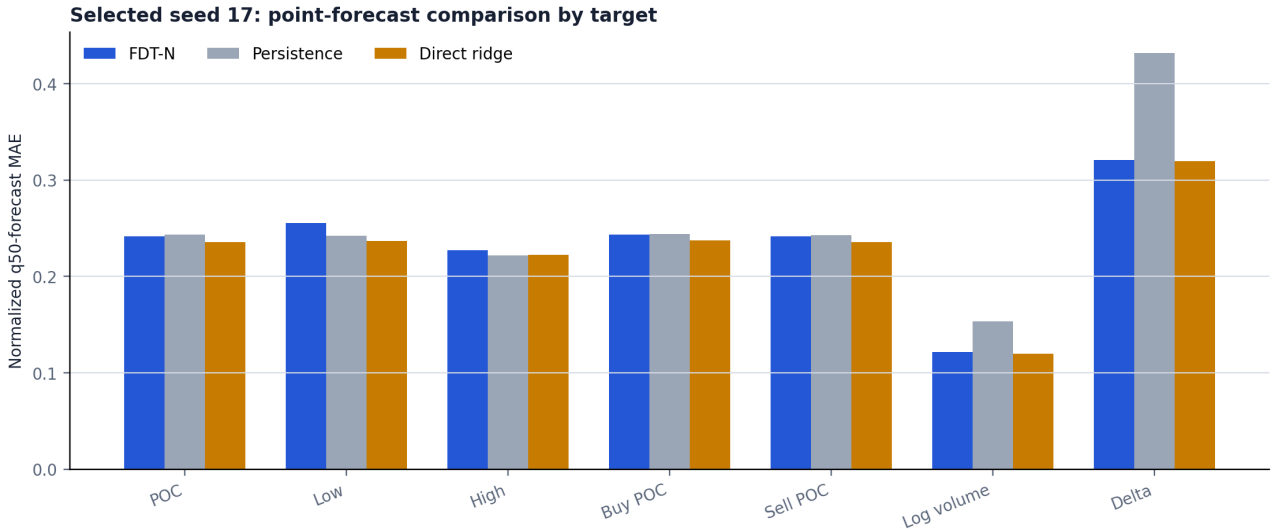


Figure 3. Selected-run q50-forecast MAE by target. Lower is better. FDT-N improves on persistence mainly for delta fraction and relative log volume, but direct ridge is lower for every target aggregate.

Target	FDT-N	Persistence	Ridge	FDT - persistence	FDT - ridge	80% coverage
POC offset	0.2414	0.2432	0.2357	-0.0017	+0.0057	78.59%
Low offset	0.2552	0.2423	0.2371	+0.0129	+0.0181	77.00%

Target	FDT-N	Persistence	Ridge	FDT - persistence	FDT - ridge	80% coverage
High offset	0.2272	0.2221	0.2225	+0.0051	+0.0047	78.87%
Buy-POC offset	0.2431	0.2442	0.2374	-0.0011	+0.0058	78.31%
Sell-POC offset	0.2415	0.2426	0.2359	-0.0011	+0.0056	80.15%
Relative log volume	0.1219	0.1536	0.1198	-0.0317	+0.0021	80.70%
Delta fraction	0.3207	0.4321	0.3197	-0.1114	+0.0011	79.23%
Pooled	0.235870	0.254298	0.229719	-0.018429	+0.006151	78.98%

Pooled FDT-N q50-forecast MAE is 7.25% lower than persistence and 2.68% higher than ridge. Excluding delta and relative-volume targets, FDT-N is worse than persistence by 0.002826 averaged across the remaining 40 horizon-target cells. The apparent pooled persistence gain is therefore not broad across price-structure targets. At cell level, ridge has lower MAE in 55 of the 56 horizon-target cells.

6.3 Point-baseline comparison by horizon

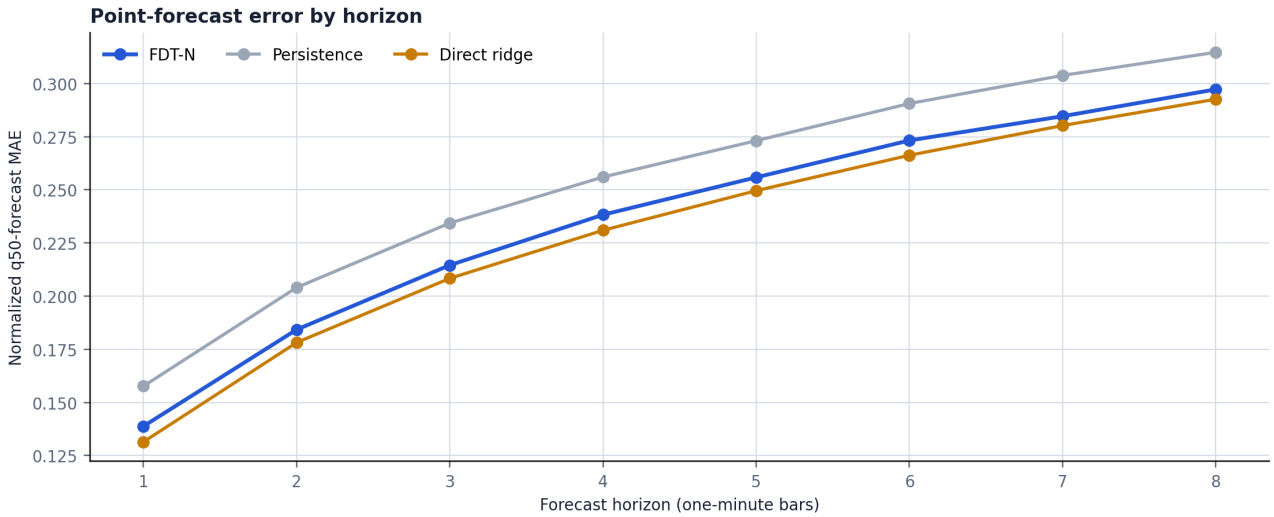


Figure 4. Q50-forecast MAE averaged over seven targets at each horizon. FDT-N is below persistence and above ridge at every horizon aggregate.

Horizon	FDT-N	Persistence	Ridge	FDT - persistence	FDT - ridge	80% coverage
1	0.1388	0.1577	0.1315	-0.0189	+0.0074	83.41%
2	0.1843	0.2041	0.1782	-0.0198	+0.0061	79.45%
3	0.2146	0.2344	0.2084	-0.0198	+0.0062	78.55%
4	0.2383	0.2560	0.2310	-0.0177	+0.0073	78.80%
5	0.2559	0.2731	0.2496	-0.0173	+0.0063	77.44%
6	0.2732	0.2906	0.2663	-0.0174	+0.0070	77.64%
7	0.2846	0.3038	0.2802	-0.0192	+0.0044	78.07%
8	0.2973	0.3147	0.2926	-0.0174	+0.0046	78.45%

6.4 Marginal coverage and resampling summaries

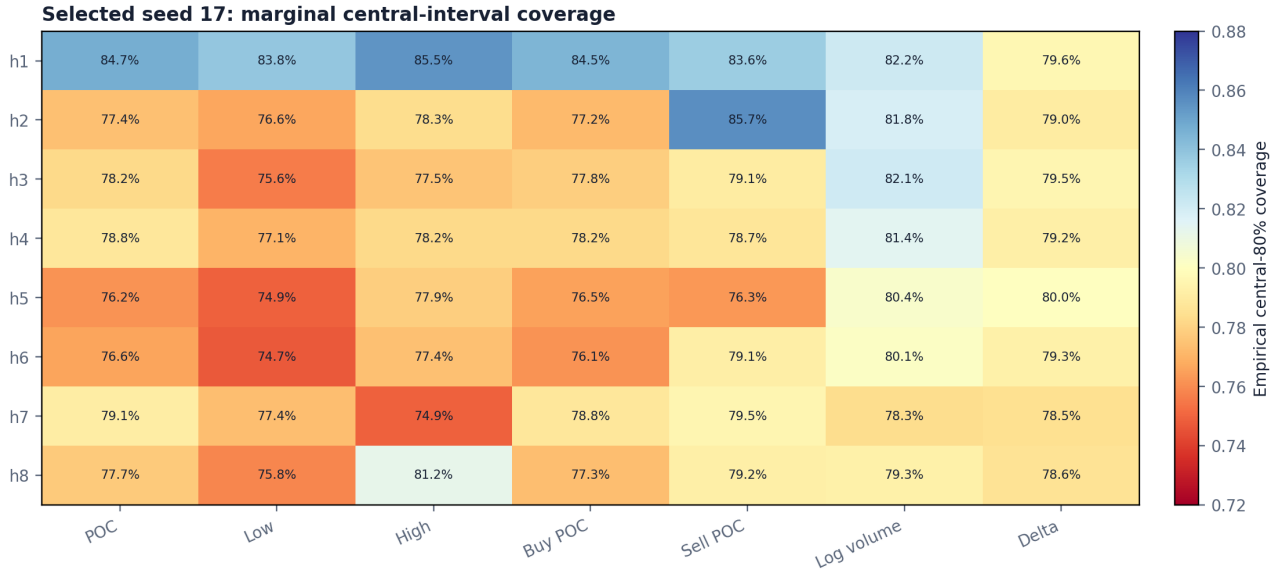


Figure 5. Empirical central-80% marginal coverage by target and horizon for seed 17. Pooled coverage is near 80%, but cellwise variation and the absence of a separate calibration set preclude a general calibration claim.

Selected-run pooled central-80% marginal coverage is 78.98%; individual horizon-target cells range from 74.75% to 85.66%. The artifacts do not record separate empirical cumulative calibration at q10, q50, and q90. Accordingly, the result is reported as central-interval coverage near nominal in aggregate, not as calibrated quantiles.

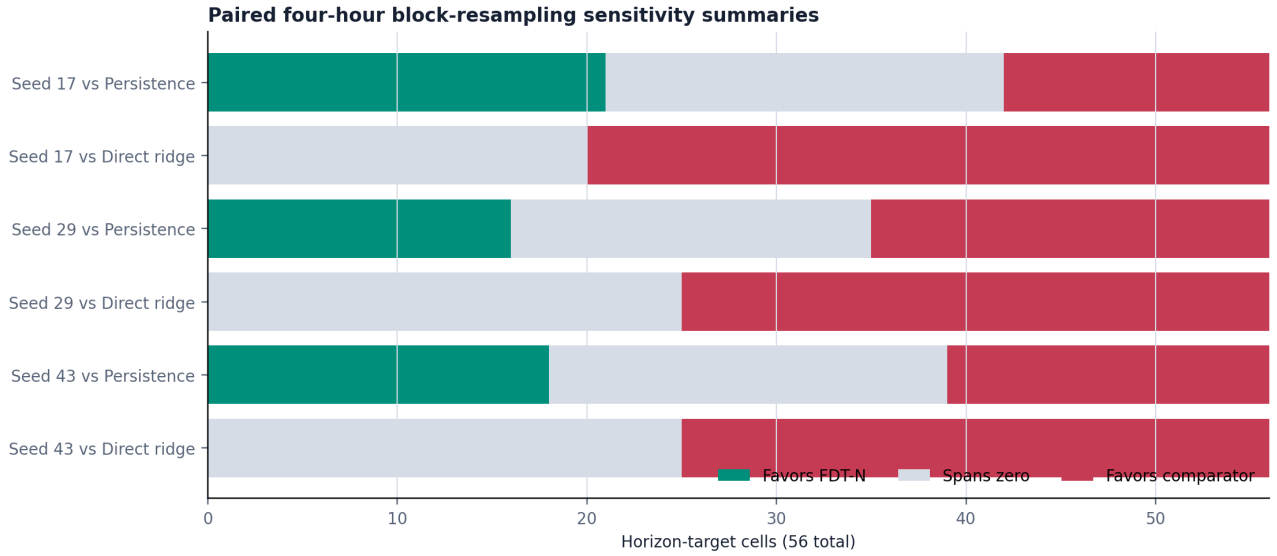


Figure 6. Counts of 56 horizon-target descriptive intervals by comparator and seed. No interval favors FDT-N over direct ridge. These dependent-block resampling summaries are descriptive, not formal confidence intervals.

Seed	Comparator	Favors FDT-N	Spans zero	Favors comparator
17	Persistence	21	21	14
29	Persistence	16	19	21
43	Persistence	18	21	17
17	Direct ridge	0	20	36
29	Direct ridge	0	25	31
43	Direct ridge	0	25	31

6.5 Clipping and runtime evidence

Price-target clipping affects approximately 1.04%-1.17% of observations, depending on target and bound. Encoded row-price offset boundary saturation is 3.51%; the frozen dataset does not retain the unclipped row coordinate needed to reconstruct exact pre-clip row loss. The reported metrics therefore apply to the clipped normalized target space.

The validation-selected checkpoint is exported to ONNX. On one representative batch-one window with 187 rows at the widest bar, ONNX Runtime CPU output differs from PyTorch by 2.38e-7 maximum and 5.17e-8 mean absolute error, below the declared 1e-4 tolerance. Fifty isolated CPU runs yield 11.410/14.641/17.240 ms p50/p95/p99 latency. This is artifact timing, not end-to-end chart latency, throughput, peak memory, operational safety, or production qualification.

7. Discussion

7.1 What the experiment supports

FDT-N shows that a variable-size executed-volume footprint can be represented without silent row compaction, processed with bounded learned-query attention, and mapped to ordered multi-horizon marginal quantiles in a compact 371,861-parameter model. Training is reproducible enough for three seeds to produce tightly grouped test pinball loss and q50-forecast MAE. The exported graph has strong cross-runtime numerical parity, and pooled central-80% coverage lies close to its nominal level.

The experiment does not show that the architecture improves forecasting over a strong simple model. Direct ridge has lower q50-forecast MAE in every target aggregate, every horizon aggregate, and 55 of 56 individual horizon-target cells; no selected-run descriptive interval favors FDT-N over ridge. This result is consistent with evidence that linear controls can be difficult to beat on structured time-series tasks [7]. It also suggests that the corpus is small relative to the model's flexibility, aggregate recency/trend features carry much of the point-prediction signal, the current objective allocates capacity to interval prediction rather than q50 alone, or the proposed row representation lacks useful incremental information. These explanations are hypotheses, not conclusions.

7.2 Why the negative ridge result matters

Reporting only the persistence comparison would produce a misleading success narrative. The persistence gain is concentrated in delta and relative volume; price extremes are worse, and total and side POC improvements are small. Ridge narrows this gap with causal aggregate features and a fraction of the model complexity. A useful next paper must therefore test whether full rows improve proper distributional scores under matched budgets, rather than assuming that richer input is beneficial.

7.3 Required next experiments

The highest-value next experiment is a preregistered factorial comparison on a later untouched period: aggregate-only versus full-row input; fixed pooling versus learned queries; ridge/quantile trees/MLP versus temporal convolution, recurrent, and transformer encoders; and point versus distributional objectives. Hyperparameters and model-selection budgets should be matched on training and validation periods. Evaluation should include a separate calibration interval, multiple market regimes, additional instruments and venues, and a dependence unit chosen before the final test.

Economic value is a separate study. Any trading experiment would require a frozen decision rule, latency model, fees, spread, slippage, fill uncertainty, position/risk constraints, and abstention policy. Forecast error, sign agreement, ONNX latency, or central-interval coverage alone cannot support a profitability or capital-allocation claim.

8. Threats to validity

External validity. The corpus contains one instrument, venue, row size, interval, and 14-day period. The test portion covers approximately two days. Results do not establish cross-market, cross-regime, seasonal, or long-run performance.

Construct validity. Footprint rows depend on trade classification, row snapping, quantity scaling, and bar sealing. Buyer-is-maker flags, inferred aggressor rules, and synthetic pseudo-prints have different error profiles. This study uses exchange-provided individual trades and does not validate synthetic flow.

Statistical validity. Forecast windows and horizons overlap. The block summaries use only 13 blocks, retain overlap across block boundaries, and lack multiplicity correction. They are descriptive sensitivity analyses. No untouched outer test or repeated chronological folds are available.

Comparator validity. Persistence and ridge provide point forecasts only. There is no matched distributional baseline, aggregate-only ablation, or architecture ablation. The study cannot attribute performance to full rows, learned queries, gated fusion, attention, or parallel horizon decoding.

Probabilistic validity. The q10/q90 outputs define marginal central intervals. Near-nominal pooled coverage does not imply conditional calibration, and constrained marginals do not form a joint candle distribution. A separate calibration set and task-matched proper-score baselines are absent.

Preprocessing validity. Target clipping and row-coordinate saturation discard tail information. Exact pre-clip row-coordinate loss cannot be reconstructed from the frozen encoded series.

Operational validity. Numerical ONNX parity and isolated CPU timing do not establish production latency, memory use, concurrency behavior, drift monitoring, or fail-safe system operation.

9. Reproducibility and artifact availability

The raw source archives are publicly available from Binance Data Vision [20]. The frozen manifest records all 14 archive identities and hashes. The processed dataset, split files, results, checkpoint, ONNX graph, metadata, and parity fixture are bound by SHA-256 in the accompanying research package. Public distribution of private implementation code and processed artifacts is not asserted by this anonymous manuscript; a submission requiring open artifacts must resolve that policy before acceptance.

Artifact	SHA-256 or identity
Source manifest	b88d89565febb97d be56151fe447e395 2fa844b8c433ecb8 6cb747e11e4ce412
Encoded dataset	d83f78894af08b65 123d2bf9f4fbef8c a28e901a552d3af2 c61f6e37513bf6cc
Seed-17 / seed-29 / seed-43 results	746403a057907539 43b8c22f615f097c 8dc397e413be49f3 5ea553b82e0cbd0a / 1b3e19b476b65e27 becb30aaf600ea03 84477b64b13f4d0d 3eb2abddf3d0e0cc / 4d28b1e41c6f93e2 3f6c86261d7710a9 974a58c5985aab7c 029415f2a4e605ad
Selected checkpoint	9d2cf525971f81fc 93ca9ed97103cffb 29203d7fd59edea5 bd92d0e1480e124a
Selected ONNX graph	adce8cd39312bf3a d1389a23ed5ca24d 6f47a7f58dbb26d7 6ae7d54af1011af7
Selected metadata	9f998ac6b56b008a c02aaa0ed5c4f084 47b1735f33027294 773d9e1aa0867763

Recorded software includes Python 3.12.6, PyTorch 2.5.1, NumPy 2.1.2, ONNX 1.22.0, and ONNX Runtime 1.28.0. Seeds, configuration, parameter count, best epoch, elapsed time, and device class are present in the result bundle. The exact GPU model, original command lines, and a clean repository revision snapshot were not frozen; these are reproducibility limitations.

10. Ethics, availability, and declarations

The study uses public market-event archives and contains no personal data or human participants. FDT-N is a research forecaster. The selected artifact is marked shadow-only and does not create orders, positions, or execution instructions. No result in this paper supports deployment, financial advice, or capital allocation.

Raw-data availability and the current artifact policy are stated in Section 9. Author identity, affiliation, ORCID, funding, competing interests, and author-contribution statements are withheld for anonymous review rather than inferred. They must be supplied, together with any venue-specific data and code statement, before non-blind publication.

11. Conclusion

FDT-N is a compact hierarchical transformer for exact-row-under-a-bound volume-footprint forecasting. It combines learned-query set pooling, gated aggregate fusion, causal temporal attention, parallel direct horizons, ordered marginal quantiles, and simple physical constraints. On a frozen 14-day BTCUSDT development corpus, its pooled q50-forecast MAE is lower than persistence but consistently higher than train-only direct ridge. Central-80% marginal coverage is near nominal in aggregate, while cellwise variation, missing calibration data, and absent distributional baselines prevent a calibration claim. The appropriate contribution is a reproducible formulation and honest negative learned-baseline result. Establishing incremental value from full footprint rows requires preregistered ablations, strong probabilistic controls, and independent multi-market replication.

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